

Association vs causation and confounding variables

MATH 213

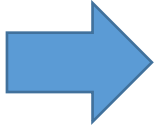
Association and Causation

Two variables are ***associated*** if values of one variable tend to be related to values of the other variable.

Two variables are ***causally associated*** if changing the value of the explanatory variable influences the value of the response variable.

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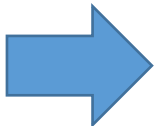


Example:

Heavy rainfall **is associated with** flooding
and

Heavy sunscreen use **is associated with**
skin cancer

Two variables are ***causally associated*** if changing the value of the explanatory variable influences the value of the response variable.

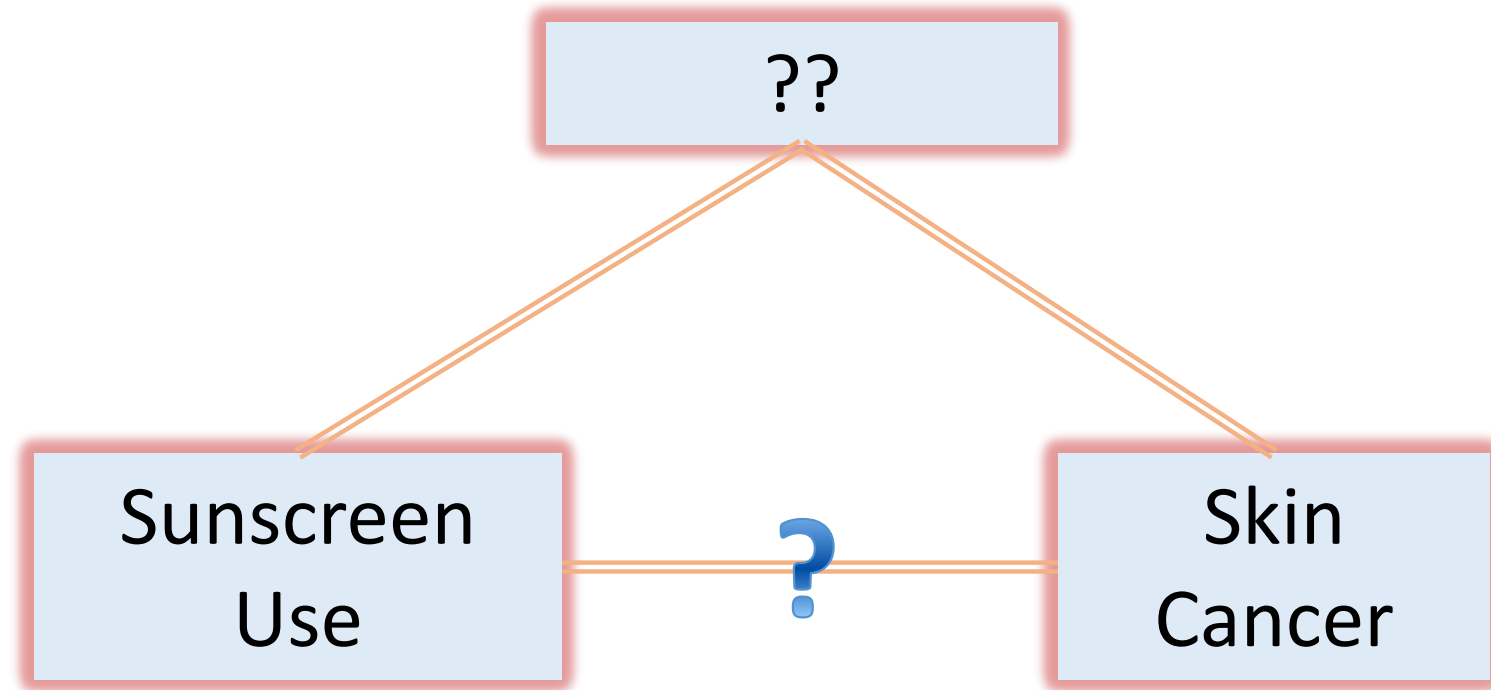


Example:

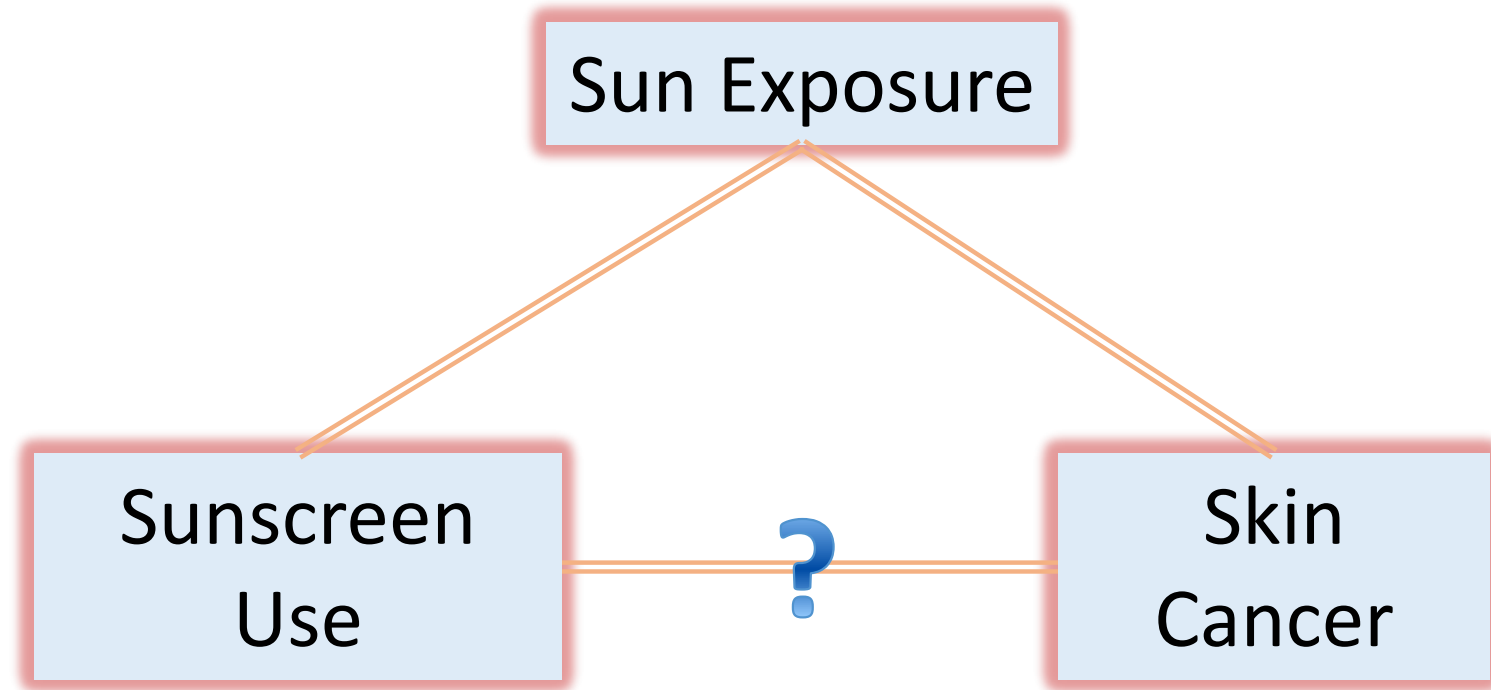
Heavy rainfall **causes** flooding
but

Heavy sunscreen use **does not cause** skin
cancer

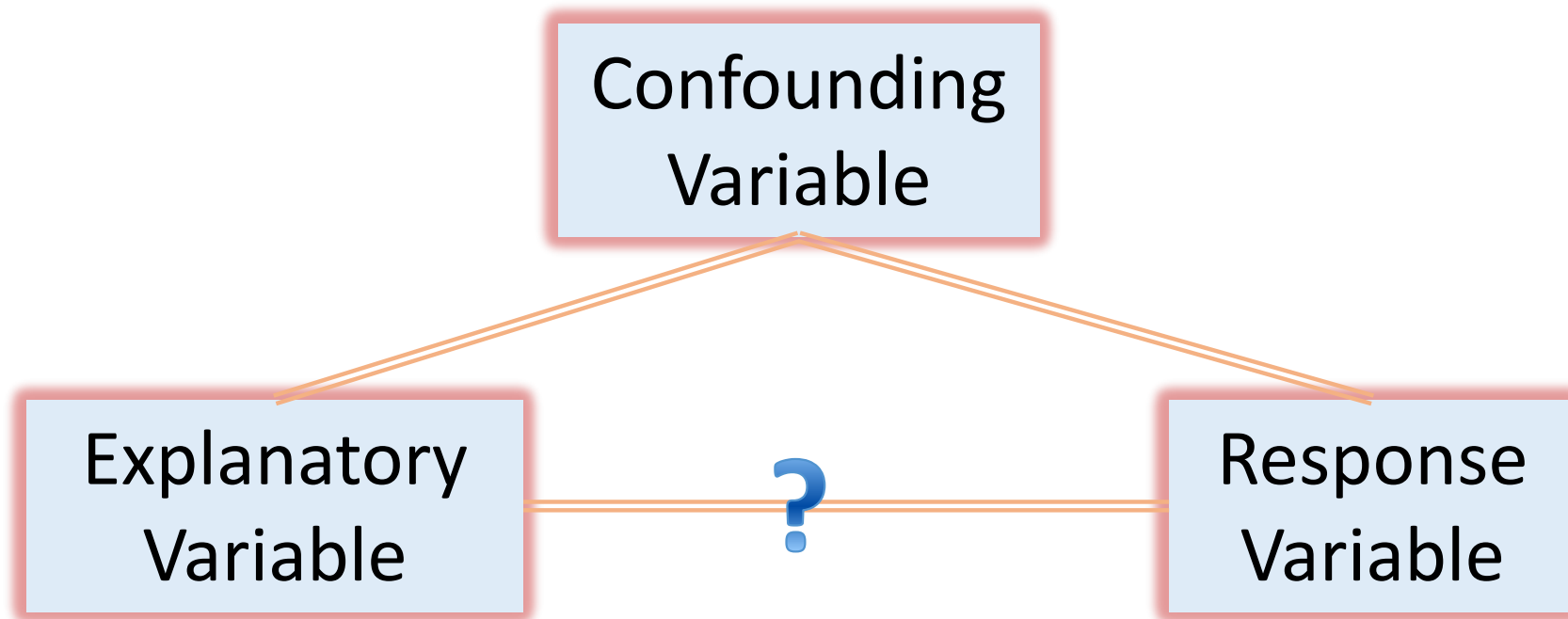
Confounding Variables



Confounding Variables



Confounding Variables



A ***confounding variable*** is a third variable that is associated with both the explanatory variable and the response variable. A confounding variable can offer a plausible explanation for an association between two variables of interest.